

Problem Statement: Agentic AI for Mental Health Awareness & Suicide Prevention

The Challenge

Mental health issues such as depression, anxiety, and suicidal thoughts are growing worldwide. Stigma, lack of awareness, and limited access to timely help prevent many individuals from seeking support. Traditional systems are often reactive, leaving gaps in early detection and preventive care.

The Objective

Build an Agentic AI-powered Mental Health Awareness & Suicide Prevention Agent that is empathetic, proactive, and safe. The solution should:

1. Promote Awareness & Education – Share accessible resources on mindfulness, coping strategies, and self-care.
2. Enable Early Detection (RAG) – Use Retrieval-Augmented Generation to identify distress signals in conversations, journaling inputs, or safe online data.
3. Provide Empathetic Support – Build a conversational agent that listens actively and responds with empathy (with clear disclaimers: not a substitute for professional medical advice).
4. Predict & Prevent Risks – Forecast stress peaks, suggest preventive strategies, and issue alerts when high-risk signals are detected (with consent).
5. Connect to Human Support – Link users to helplines, counselors, or community support for timely intervention.

Tech Stack (Mandatory)

- LangFlow + IBM watsonx.ai + IBM Granite Models

Proposed Solution:

The proposed solution is a **multi-agent AI-powered Mental Health Awareness & Suicide Prevention System** built using **LangFlow**, **IBM watsonx.ai**, and **IBM Granite Models** to provide empathetic, proactive, and personalized mental health support. The system addresses challenges such as lack of awareness, delayed intervention, and limited access to timely help through an intelligent and modular architecture.

At its core, the solution utilizes a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve reliable information from trusted mental health resources, mindfulness guides, coping strategy documents, and crisis support materials. The retrieved knowledge is embedded and stored in a vector database, enabling Granite models to generate accurate and context-aware responses.

The system consists of three key agents:

- **MindGuide AI (Awareness & Education Agent)** provides information on stress, anxiety, depression, mindfulness, self-care practices, and healthy coping strategies while promoting mental health awareness and reducing stigma.
- **Sentinel AI (Risk Detection Agent)** analyzes user conversations and journaling inputs to identify emotional distress, detect early warning signs of anxiety, depression, or suicidal ideation, classify risk levels, and recommend preventive actions using RAG-based reasoning.
- **HopeBridge AI (Empathetic Support Agent)** offers compassionate conversations, personalized coping strategies, wellness plans, and motivational guidance while encouraging users to seek professional help or connect with helplines when high-risk situations are detected.

The solution provides an interactive chat interface where users can ask questions, record journal entries, and receive personalized support. With user consent, the system can monitor emotional patterns, forecast stress peaks, and generate preventive recommendations to improve overall well-being.

LangFlow orchestrates the multi-agent workflow, **IBM watsonx.ai** provides scalable AI infrastructure and governance, while **IBM Granite Models** power retrieval, reasoning, summarization, and empathetic response generation.

Overall, the proposed solution delivers an intelligent mental health assistant that promotes awareness, enables early detection of emotional distress, provides empathetic support, predicts potential risks, and connects users to appropriate human assistance, fostering proactive and preventive mental healthcare.

Langflow component used:

1. **Chat Input** – Allows users to enter mental health queries or journal entries for AI processing.
2. **IBM watsonx.ai** – Processes user input using IBM Granite models to generate intelligent and context-aware responses.
3. **IBM Granite-4.0-h-small Model** – Provides fast and efficient reasoning for mental health awareness and conversational support.
4. **MindGuide AI Agent** – Delivers mental health education, self-care tips, and awareness-based guidance.
5. **Sentinel AI Agent** – Detects emotional distress and assesses potential mental health risks from user conversations.
6. **HopeBridge AI Agent** – Offers empathetic support, coping strategies, and encourages professional help when needed.
7. **Read File Component** – Reads the uploaded mental health PDF and provides knowledge for AI responses.
8. **Prompt Template Component** – Dynamically combines user queries with document content to create context-aware prompts.
9. **Language Model Component** – Generates responses based on the prepared prompt and retrieved knowledge.
10. **Chat Output** – Displays the final personalized mental health response to the user.

Langflow workflow

